FLASH CARD 1

Slopes of Two Lines

Two lines lie in the **same plane**.

Their slopes are m_1 and m_2.

What are the **3 cases**?

What does the sign $\perp$ mean?

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FLASH CARD 2

Writing a Parallel or Perpendicular Line

Write the equation of the line through $(2,\ 1)$ that is

(a) parallel to $y = 2x + 5$ **(b)** perpendicular to $y = 2x + 5$.

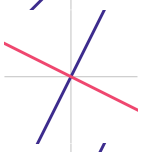
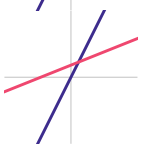
Give your answers in slope-intercept form, $y = mx + b$.

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Slopes of Two Lines — 3 Cases

Compare the slopes m_1 and m_2 of two lines in the same plane:

Picture	Slopes	The lines are…	Example
	$m_1 = m_2$	**parallel**	$y = 3x - 2$ and $y = 3x + 4$ same slope 3
	$m_1 \cdot m_2 = -1$ opposite reciprocals	**perpendicular** ($\perp$)	$y = 2x + 1$ and $y = -\frac{1}{2}x + 4$ $2 \cdot \left(-\frac{1}{2}\right) = -1$
	neither	**intersect, but not perpendicular**	$y = 2x$ and $y = x + 3$ $2 \neq 1$, and $2 \cdot 1 = 2 \neq -1$

$\perp$ is the **perpendicular sign**: $\ell_1 \perp \ell_2$ means "line ℓ_1 is perpendicular to line ℓ_2" (they meet at a right angle).

Watch out: (1) Same slope *and* same y-intercept means the *same line*, not two parallel lines. (2) A vertical line ($x = a$) and a horizontal line ($y = b$) are perpendicular, even though a vertical line has no slope, so $m_1 \cdot m_2 = -1$ cannot be used.

Card 1 · Answer

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Writing a Parallel or Perpendicular Line

Steps: **1.** Find the slope you need. **2.** Use point-slope form $y - y_1 = m(x - x_1)$. **3.** Solve for y.

(a) Parallel

Same slope as $y = 2x + 5$: $m = 2$.

$$\begin{aligned} y - 1 &= 2(x - 2) \\ y - 1 &= 2x - 4 \\ y &= 2x - 3 \end{aligned}$$

(b) Perpendicular

Opposite reciprocal of 2: $m = -\frac{1}{2}$.

$$\begin{aligned} y - 1 &= -\tfrac{1}{2}(x - 2) \\ y - 1 &= -\tfrac{1}{2}x + 1 \\ y &= -\tfrac{1}{2}x + 2 \end{aligned}$$

Check: put in $x = 2$. (a) $2 \cdot 2 - 3 = 1$ ✓ (b) $-\frac{1}{2} \cdot 2 + 2 = 1$ ✓

Common Mistake: stopping at $y - 1 = 2x - 4$, or dropping the -1 and writing $y = 2x - 4$. Finish the last step: add 1 to *both* sides so y is alone.

Card 2 · Answer

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